

GROWTH INTERN ASSIGNMENT

Jar — Growth Intern Assignment

Sales analysis · App exploration · Product exploration

[Candidate name] · 9 October 2026

Code: [GitHub repository link]

Analysis in Python · all figures and numbers regenerated by `python main.py`

Contents

Executive summary	3
1. Data and method	3
1.1 Join validation	3
1.2 Date repairs	3
1.3 Definitions	4
1.4 Sample size	4
2. Q1 Part 1 — Sales and profitability	4
2.1 Category scorecard	4
2.2 Why the categories differ	5
3. Q1 Part 2 — Target achievement (Furniture)	8
3.1 Month-over-month change in target	8
3.2 Target vs actual	9
3.3 Aligning targets with performance	9
4. Q1 Part 3 — Regional performance	10
4.1 Top 5 states by order count	10
4.2 Regional disparities	11
4.3 Where to focus	12
5. Q2 — App exploration	13
5.1 Five things that work well	13
5.2 Five areas to improve	13
6. Q3 — Product exploration	15
6.1 Starting point: Jar's strengths and the category's open gap	15
6.2 Opportunity cards	16
6.3 Sequenced roadmap	17
6.4 Sources (Sections 5–6)	17
Appendix A — Assumptions log	18
Appendix B — Detail tables	19
Appendix C — Reproducibility	22
Appendix D — Q2 hands-on checklist (≈ 20 minutes)	23

Executive summary

1. **Categories.** Electronics leads on sales (₹1,65,267, 38.3%) and profit per order (₹51.44). Clothing earns the most profit (₹11,163) at the best margin (8.03%). Furniture trails on every metric (1.81% margin).
2. **Profit leaks.** Two sub-categories lose money: Tables (-₹4,011) and Electronic Games (-₹1,236). In 16 of 17 sub-categories, loss-making lines realise lower unit prices than profitable ones (median 0.55×) at the same quantity.
3. **Furniture targets.** The target ramps smoothly (no MoM move above 1.89%), but demand is back-loaded. Furniture missed target 7 months in a row, then beat it in 4 of the last 5 (95.7% for the year). Seasonal phasing cuts monthly error from 54.4% to 19.5%.
4. **Regions.** Madhya Pradesh and Maharashtra hold 38.2% of orders. Mumbai's margin is 2.65%, against 13.56% in Pune. 4 states lose money (-₹3,642), while Uttar Pradesh, Delhi, West Bengal and Kerala earn 44.2% of profit on 16.8% of sales.
5. **Product (Q3).** Digital gold sits outside SEBI's oversight industry-wide (advisory, November 2025). Jar can lead the category by setting the disclosure standard and moving its automation onto regulated rails. The first moves are Nek goal savings and a regulated gold SIP, followed by insured deposits, sachet insurance and partner-led gold credit.

1. Data and method

1.1 Join validation

Order Details (one row per line item) is **left-joined** to List of Orders on Order ID with pandas `validate="many_to_one"` and an indicator column. Order Details carries the grain of every financial metric. The many-to-one check proves Order ID is unique on the header side, so the join cannot duplicate lines. A hard assertion fails the run on any orphan, instead of letting an inner join drop it silently.

Check	Result
List of Orders rows / unique Order IDs	500 / 500
Order Details rows / unique Order IDs	1,500 / 500
Duplicate Order IDs in List of Orders	0
IDs only in List of Orders / only in Order Details	0 / 0
Rows after merge	1,500 (equals line items)

Orders have 1 to 12 lines, and 207 span more than one category. That is why "per order" and "per line" averages differ.

1.2 Date repairs

Spreadsheet auto-conversion corrupted both date fields. Both were repaired in code, with assertions; the raw files are untouched.

- **Order Date.** 307 cells are `dd-mm-yyyy` text. The other 193 were read month-first, so day and month were swapped back. Repaired dates rise monotonically with the sequential Order ID and span 01 Apr 2018 to 31 Mar 2019.

- **Sales Target month.** "Apr-18" had been stored as *18 April of the current year*, so year = 2000 + stored day. The months run April 2018 to March 2019, matching the order period exactly.

1.3 Definitions

- **Sales** = sum of Amount. **Margin** = $\text{sum(Profit)} \div \text{sum(Amount)}$. **Order count** = distinct Order IDs.
- **Average profit per order (primary)** = $\text{group profit} \div \text{distinct orders containing the group}$. **Per line item (secondary)** = mean line profit. Both rank the categories identically; per-order values are 1.3×–2.4× higher.
- **Price realisation.** The data has no discount field, so discounting can only be tested indirectly: realised unit price ($\text{Amount} \div \text{Quantity}$) on loss-making lines is compared with profitable lines in the same sub-category.

1.4 Sample size

The dataset is small: 500 orders and 1,500 lines over one year. Several cuts rest on few observations: Tables has 17 lines, Tamil Nadu 8 orders, and Furniture sells 10–32 orders a month. A handful of large lines can move a sub-category, city or month. Category-level findings are robust. Single-city and single-month results are directional and should be confirmed with more data before acting.

2. Q1 Part 1 — Sales and profitability

2.1 Category scorecard

Electronics leads on sales and profit per order; Furniture trails on every metric

Category totals across all 500 orders, Apr 2018 - Mar 2019

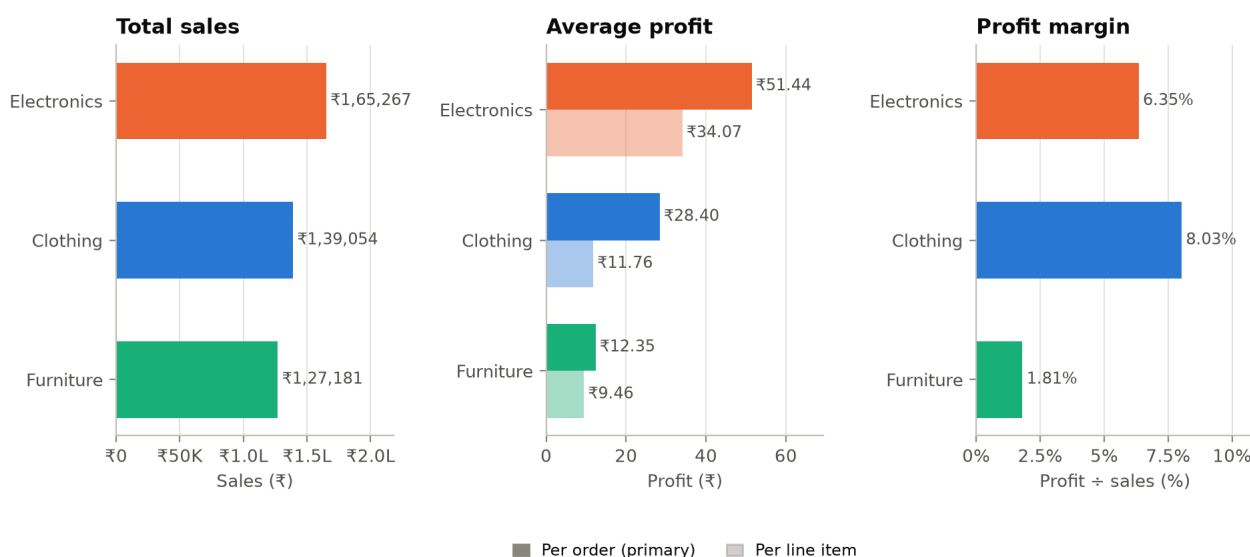


Figure 1. Category totals: sales, average profit (per order and per line) and margin.

Category	Sales	Share of sales	Profit	Share of profit	Profit / order	Profit / line	Margin
Electronics	₹1,65,267	38.3%	₹10,494	43.8%	₹51.44	₹34.07	6.35%
Clothing	₹1,39,054	32.2%	₹11,163	46.6%	₹28.40	₹11.76	8.03%
Furniture	₹1,27,181	29.5%	₹2,298	9.6%	₹12.35	₹9.46	1.81%
Total	₹4,31,502	100%	₹23,955	100%			5.55%

Verdict.

- **Electronics leads on scale and order economics:** #1 on sales and on profit per order.
- **Clothing leads on efficiency:** #1 on total profit and on margin. Its per-line profit (₹11.76) understates it, because its lines are small (unit price ₹39.55). It also appears in 393 of 500 orders, the most of any category.
- **Furniture underperforms:** last on all four metrics.

2.2 Why the categories differ

1. One or two sub-categories hold each category back.

Two sub-categories lose money: Tables (-₹4,011) and Electronic Games (-₹1,236)

Total profit by sub-category; bars sorted within each category

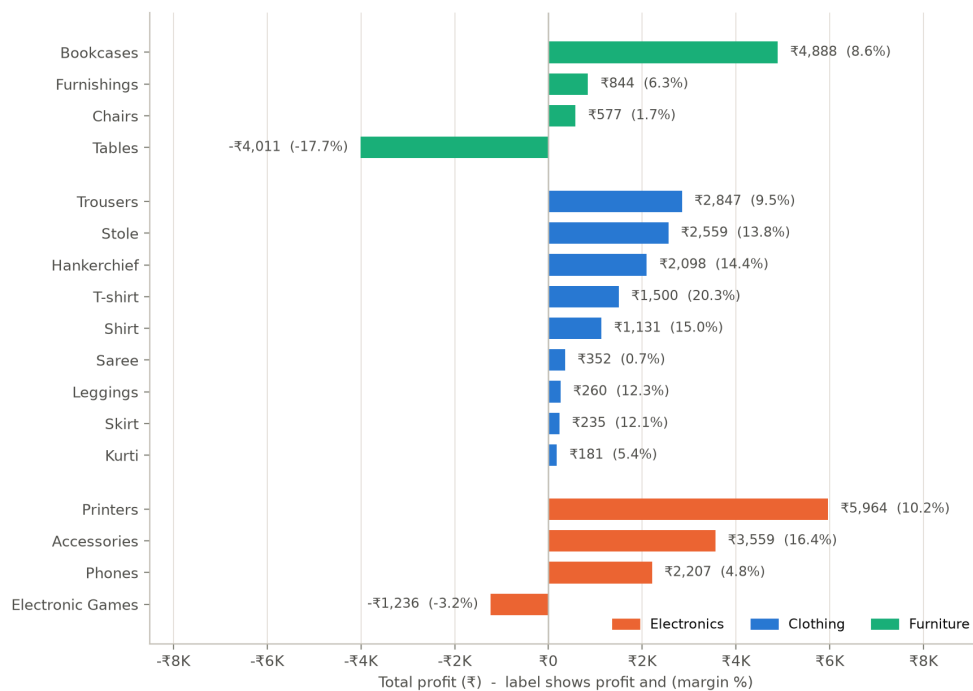


Figure 2. Profit and margin by sub-category.

- **Furniture.** Tables lose -₹4,011 on ₹22,614 (-17.7% margin; 64.7% of lines in loss). Chairs barely break even (1.7%). Bookcases earn ₹4,888, which is 213% of Furniture's net profit. Without Tables, the margin would be 6.03%, close to Electronics (6.35%).
- **Electronics.** Electronic Games lose -₹1,236 (-3.2%), and Phones earn 4.8% with 49.4% of lines in loss. Printers (56.8% of category profit) and Accessories (16.4% margin) carry the category. Without Games, the margin would be 9.30%.
- **Clothing.** Sarees are 38.5% of sales at a 0.66% margin. The rest of Clothing earns 12.6%.

2. Losses are concentrated, most severely in Furniture.

Furniture's loss-making lines erase 87% of what its profitable lines earn

Line-level gains and losses by category

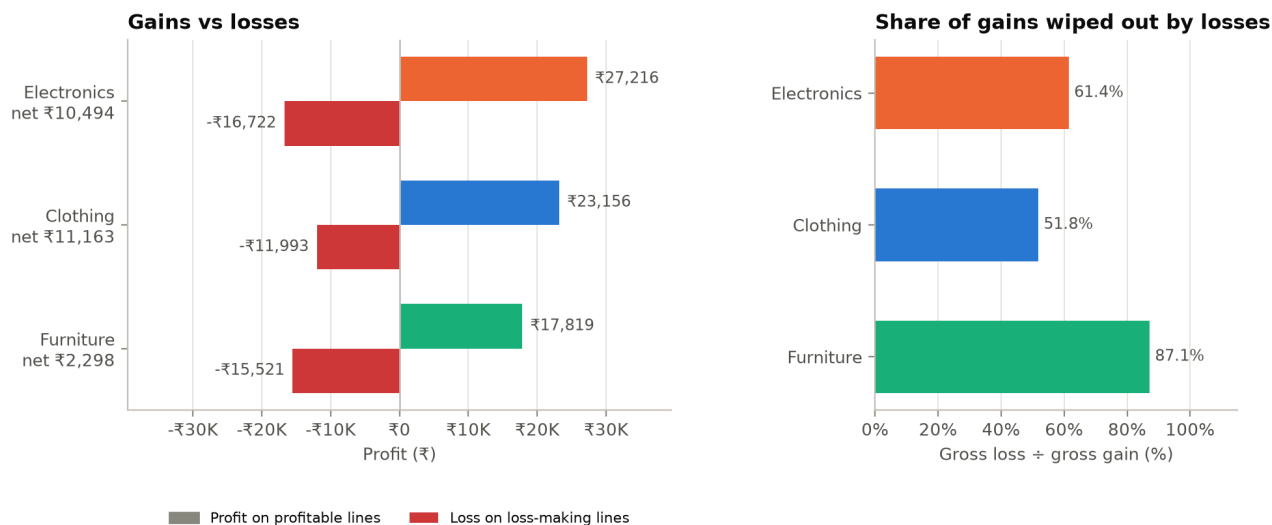


Figure 3. Gross gains from profitable lines vs gross losses from loss-making lines.

Category	Lines in loss	Loss ÷ gain	Worst 10 lines' share of losses	Lines losing more than their revenue
Electronics	39.6%	61.4%	41.5%	3
Clothing	29.4%	51.8%	28.7%	1
Furniture	42.0%	87.1%	57.6%	10

Furniture's loss-making lines give back ₹15,521 of the ₹17,819 its profitable lines earn. Its worst 10 lines account for 57.6% of those losses.

3. Loss-making lines realise lower prices, not smaller quantities.

In 16 of 17 sub-categories, loss lines sell at a lower unit price (median 0.55x)

Price-realisation check: no discount field exists, so realised unit price (Amount ÷ Quantity) is compared

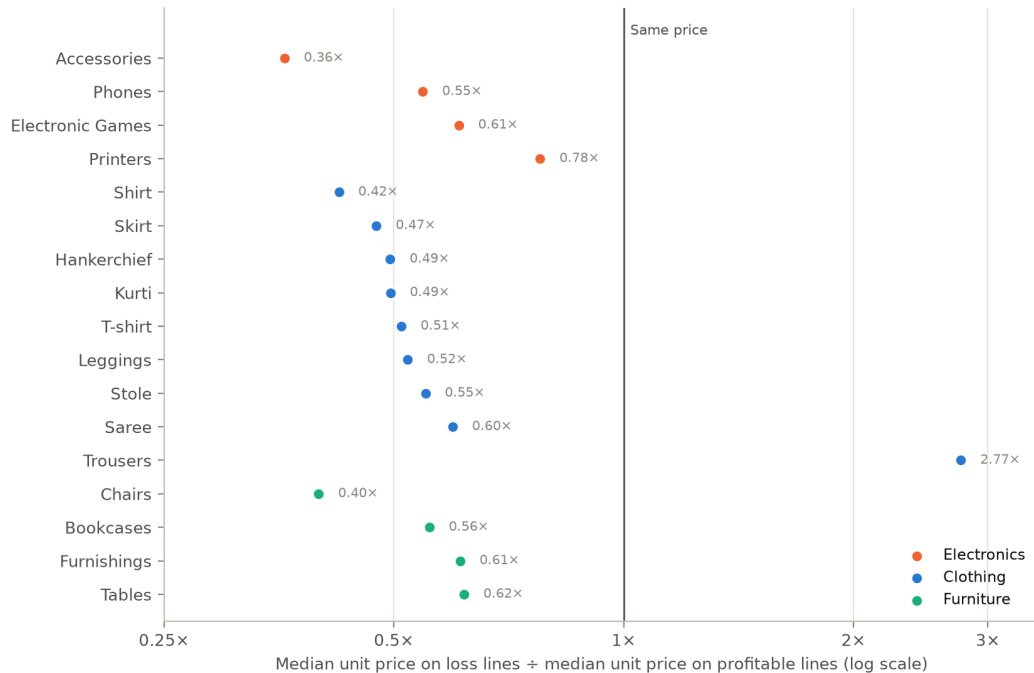


Figure 4. Median unit price on loss-making vs profitable lines within each sub-category.

In 16 of 17 sub-categories, loss-making lines sell at a lower median unit price than profitable ones (median ratio 0.55x; Chairs 0.40x). The exception is Trousers (2.77x), where losses sit on high-ticket lines, which suggests a cost problem. Median quantity is the same on both (3 units), so the losses track **low realised prices**, not volume.

The data cannot say *why* prices are low. Discounting (the same item sold for less) is one hypothesis. **Product mix** is the alternative: cheaper variants within a sub-category may carry thin or negative margins. Separating the two needs SKU-level list prices.

4. Furniture keeps little profit per unit. Furniture and Electronics sell at similar unit prices (₹134.58 vs ₹143.21). Furniture keeps ₹2.43 per unit against ₹9.09 for Electronics, so a small price concession tips a Furniture line into loss.

Actions.

1. Set price floors on Tables and Chairs.
2. Re-price or delist Electronic Games.
3. Set a floor on Saree realised prices.
4. Pull SKU data to test discount versus mix.

3. Q1 Part 2 — Target achievement (Furniture)

3.1 Month-over-month change in target

The Furniture target ramps smoothly: every MoM change is between 0.87% and 1.89%

Red = flagged by the z-score rule ($|\text{MoM \%} - \text{mean}| > 1.5 \text{ SD}$)

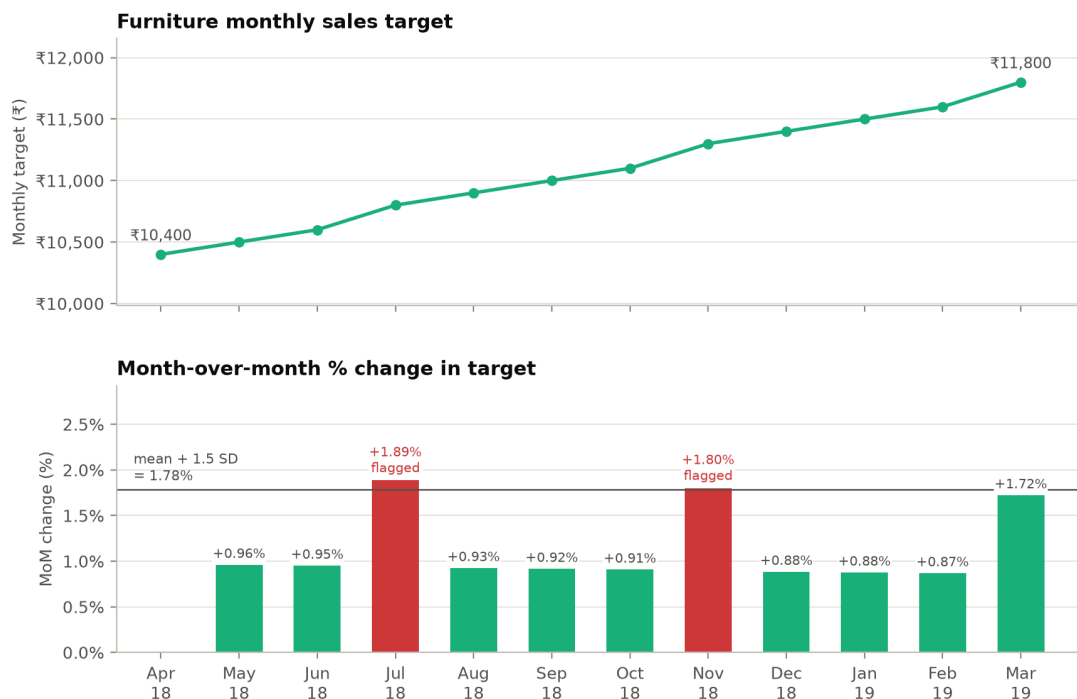


Figure 5. Furniture monthly target and its month-over-month % change.

Fluctuation rule. A month is flagged when its MoM % lies more than 1.5 standard deviations from the series mean (z-score; threshold 1.78%). As a materiality cross-check, it is also flagged if $|\text{MoM}| \geq 5\%$.

Month	Target	Change (₹)	MoM %	z-score	Flag (z-rule)
Apr 2018	₹10,400	—	—	—	
May 2018	₹10,500	+₹100	+0.96%	-0.46	
Jun 2018	₹10,600	+₹100	+0.95%	-0.48	
Jul 2018	₹10,800	+₹200	+1.89%	+1.74	Flagged
Aug 2018	₹10,900	+₹100	+0.93%	-0.55	
Sep 2018	₹11,000	+₹100	+0.92%	-0.57	
Oct 2018	₹11,100	+₹100	+0.91%	-0.59	
Nov 2018	₹11,300	+₹200	+1.80%	+1.54	Flagged
Dec 2018	₹11,400	+₹100	+0.88%	-0.65	
Jan 2019	₹11,500	+₹100	+0.88%	-0.66	
Feb 2019	₹11,600	+₹100	+0.87%	-0.68	
Mar 2019	₹11,800	+₹200	+1.72%	+1.36	

The target rises 13.5% across the year, in steps of ₹100, with ₹200 steps in Jul 2018, Nov 2018 and Mar 2019. The z-rule flags **Jul 2018 and Nov 2018**, with Mar 2019 just under the bar. The materiality rule flags **0 months**. The target's fluctuations are statistically visible but commercially trivial. The problem is a target *too smooth* for a volatile business.

3.2 Target vs actual

Flat targets miss a back-loaded year: 7 straight misses, then 4 of the last 5 months above target

Actual = sum of Furniture line Amounts per repaired order month

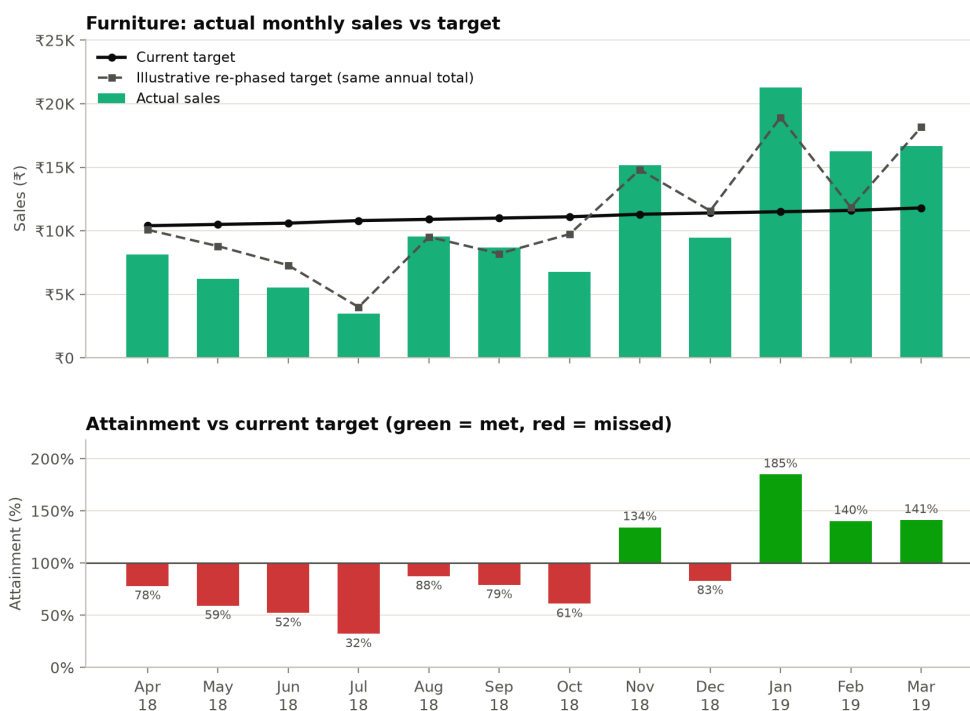


Figure 6. Actual Furniture sales vs current and illustrative re-phased target, and monthly attainment.

- **Annually about right:** ₹1,27,181 achieved against ₹1,32,900 (95.7%).
- **Monthly, phased wrong.** Target was met in only 4 of 12 months. Furniture missed for 7 straight months to Oct 2018 (64.2% attainment, -₹26,936). It then beat target in 4 of the last 5 (136.8%, +₹21,217).
- **H1 over-weighted.** April–September carries 48.3% of the target but only 32.7% of actual sales.
- **Actuals are 12.5× more volatile than the target.** The coefficient of variation is 51.3% for actuals vs 4.1% for the target.
- **Revenue-only targets hide losses.** Furniture lost money in 6 months (-₹9,316 in total).

3.3 Aligning targets with performance

1. **Phase by seasonality.** Spread the same ₹1,32,900 across months by each month's share of company-wide sales. The monthly target then ranges from ₹3,993 (Jul 2018) to ₹18,923 (Jan 2019). MAPE falls from 54.4% to 19.5%, and months within $\pm 15\%$ of target rise from 1 to 6.
 - *Caveats.* Leakage is only partial: company sales include Furniture itself (29.5% of the total), and seasonality comes from the same year, so this is an upper bound. With Clothing and Electronics alone, which excludes Furniture entirely, MAPE is 27.3%. In practice, phase on prior-year or rolling three-year shares.
2. **Add a margin guardrail** (e.g. margin $\geq 0\%$ each month), and tie price approvals on Tables and Chairs to it.

3. **Use a $\pm 15\%$ band and re-forecast quarterly.** With an actual CV of 51%, a point target will be missed almost every month.
4. **Re-balance across categories.** Electronics hit 128.1% of target and Clothing 79.9%. Shift ambition toward demonstrated run-rates.

4. Q1 Part 3 — Regional performance

4.1 Top 5 states by order count

Two states hold 38.2% of orders; Punjab makes the top 5 yet loses money

Top 5 states by distinct Order ID count. Red = negative

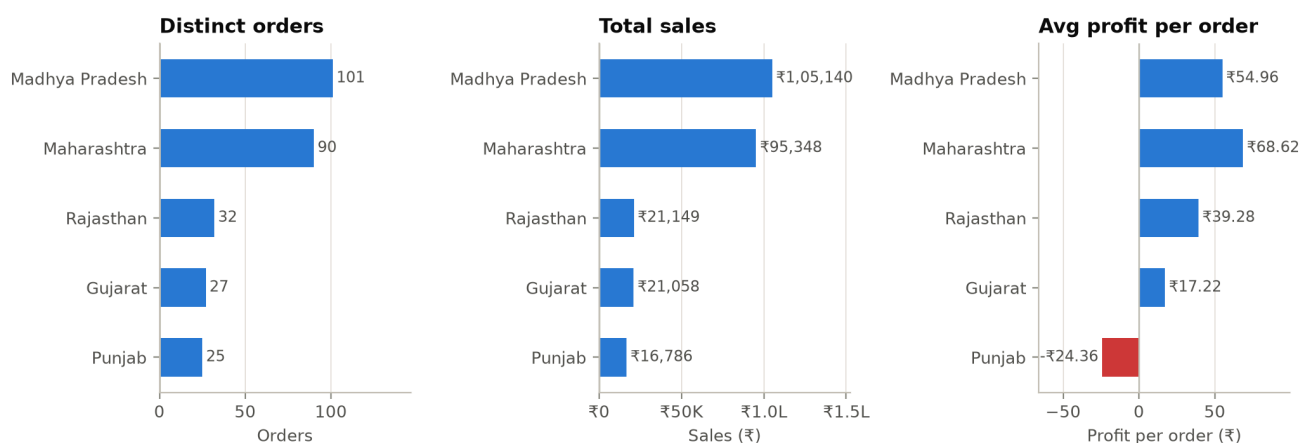


Figure 7. Top 5 states by distinct orders: order count, total sales and average profit per order.

Rank	State	Distinct orders	Total sales	Total profit	Avg profit / order	Avg profit / line	Margin
1	Madhya Pradesh	101	₹1,05,140	₹5,551	₹54.96	₹16.33	5.28%
2	Maharashtra	90	₹95,348	₹6,176	₹68.62	₹21.30	6.48%
3	Rajasthan	32	₹21,149	₹1,257	₹39.28	₹16.99	5.94%
4	Gujarat	27	₹21,058	₹465	₹17.22	₹5.34	2.21%
5	Punjab	25	₹16,786	-₹609	-₹24.36	-₹10.15	-3.63%

Average profit is per order (state profit $\div$ distinct orders); per-line is shown for reference. There is no tie at the cut-off: the next states (Delhi, Uttar Pradesh and West Bengal) have 22 orders each. The top 5 hold 55.0% of orders and 60.1% of sales, but only 53.6% of profit.

4.2 Regional disparities

4 states lose money; 4 smaller states earn well above their sales share

Blue = state captures more profit than its sales share; red = less

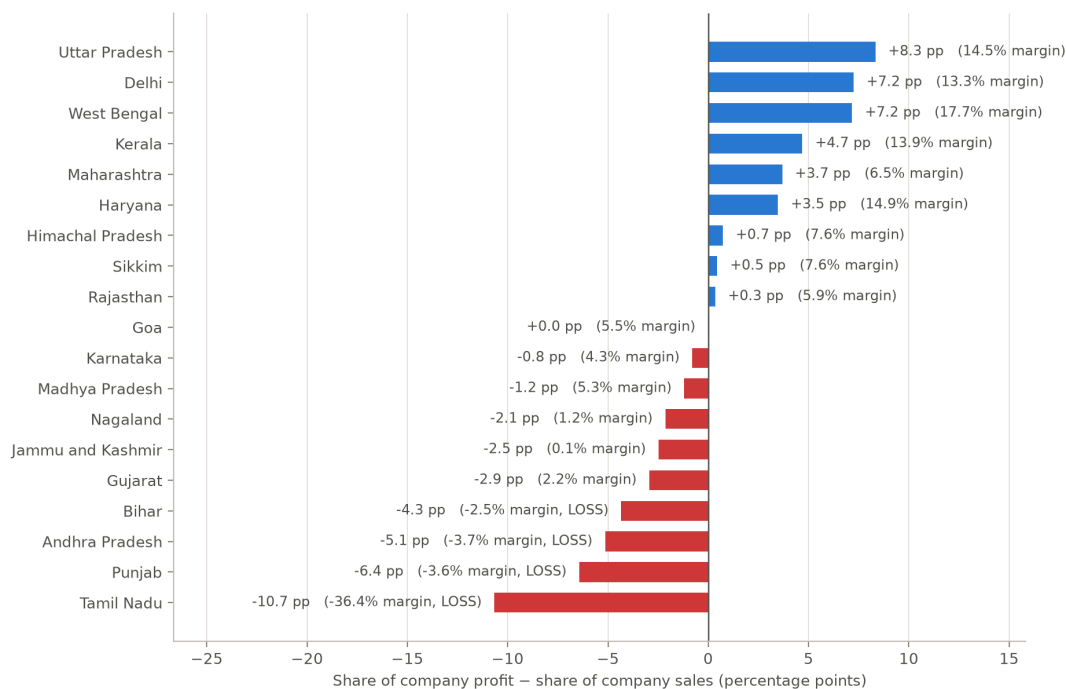


Figure 8. Each state's share of company profit minus its share of company sales.

- **Sales-to-profit mismatch.** Madhya Pradesh has 24.4% of sales but 23.2% of profit. Uttar Pradesh, Delhi, West Bengal and Kerala earn 44.2% of profit on 16.8% of sales, a 14.6% margin against 5.55% for the company.
- **Loss-making states, each driven by one category.**
 - Tamil Nadu: -₹2,216, from Furniture.
 - Punjab: -₹609, from Electronics (-₹2,132).
 - Andhra Pradesh: -₹496, from Furniture.
 - Bihar: -₹321, from Electronics.
- **City-level splits.** In Gujarat, Punjab and Rajasthan, one city makes money and the other loses it:

State	Loss-making city	Sales	Profit	Margin	Profitable city	Margin
Gujarat	Ahmedabad	₹14,230	-₹880	-6.2%	Surat	19.7%
Punjab	Chandigarh	₹12,279	-₹1,153	-9.4%	Amritsar	12.1%
Rajasthan	Jaipur	₹10,076	-₹753	-7.5%	Udaipur	18.2%

Mumbai and Indore carry the volume at thin margins; Chennai is the deepest loss

Each bubble is a city (size = distinct orders). Labels show priority cities and their margin

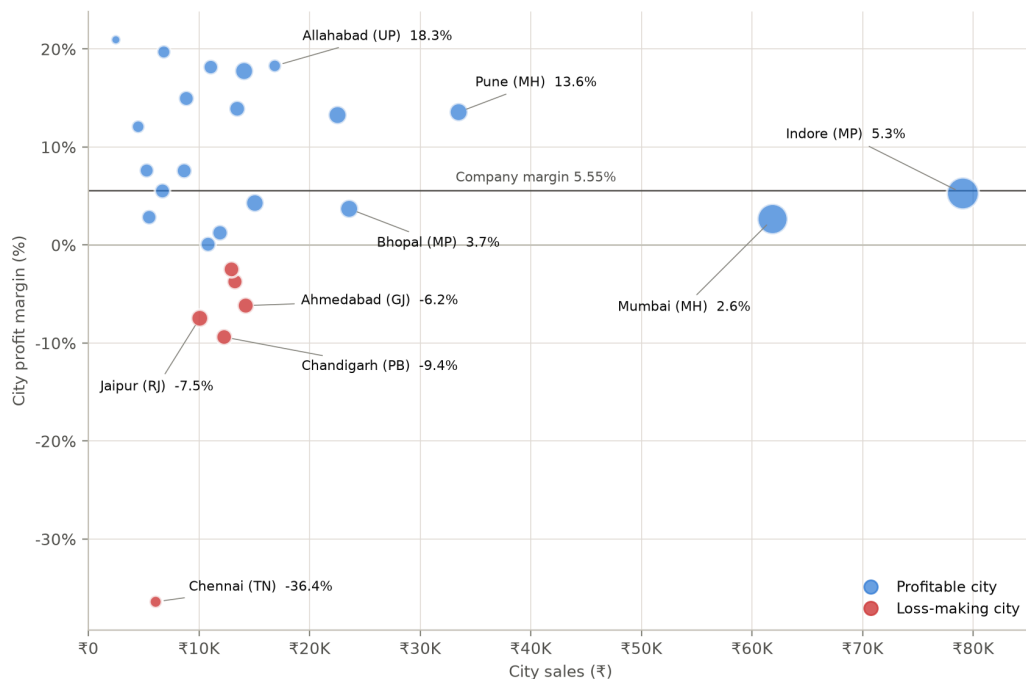


Figure 9. City sales vs margin; bubble size is distinct orders.

4.3 Where to focus

Priority	Where	Why (number)	Lever
1. Margin repair at scale	Mumbai	₹61,867 at 2.65%; 41.1% of lines in loss vs 28.9% in Pune	Pune's margin would add about ₹6,750; the company average, ₹1,798
2. Investigate	Chennai (Tamil Nadu)	-₹2,216 on ₹6,087 (-36.4%) from only 8 orders	Find what drives the Furniture loss (-₹2,487): pricing, freight or one-off returns. Act once the cause is confirmed
3. Fix loss cities in the top 5	Ahmedabad, Chandigarh and Jaipur	-₹2,786 combined; sister cities earn double-digit margins	Copy the sister city's mix and pricing; for Punjab, start with Electronics
4. Defend the volume base	Indore, Bhopal	Indore is 75.2% of Madhya Pradesh sales; Bhopal earns 3.7%	Hold Indore near the company margin; lift Bhopal
5. Grow where profit is easy	Allahabad, Delhi, Kolkata, Thiruvananthapuram	14.6% combined margin on only 82 orders	Spend acquisition budget here first

Data note: 3 orders list city "Delhi" under Madhya Pradesh; they are kept as recorded and change no ranking. Chandigarh appears under both Punjab (16 orders) and Haryana (14).

5. Q2 — App exploration

How to read this section. This is a *pre-session framework* built only from Jar's public material (website, store listing, press). Every observation tagged **[VERIFY]** must be confirmed, corrected or replaced from my own hands-on session before submission, and each has a screenshot slot. The checklist is in Appendix D.

Context from public sources. Jar reports more than 35 million registered users across 12,000 PIN codes. About 60% are in tier-2 and tier-3 towns, and more than 95% are first-time formal savers (TechCrunch, Sept 2025). Jar says the app supports nine Indian languages. Its site lists daily (from ₹10), weekly (from ₹50) and monthly (from ₹100) automated saving, Brink's as vault partner, ICICI Lombard insurance and Vistra as independent auditor/trustee.

5.1 Five things that work well

#	What works	Why it works (UX principle)	Evidence to capture
1	₹10 daily saving on UPI AutoPay. A one-time mandate replaces a daily decision.	<i>Habit loop and default effect.</i> Saving becomes the default, so the cue, routine and reward run without willpower. ₹10 removes affordability friction for first-time savers.	[VERIFY] Taps from install to first AutoPay mandate; is ₹10 the pre-selected amount?
2	Trust signals placed where money moves. Vault partner, insurer, trustee audit, 99.9% purity.	<i>Trust signals in fintech.</i> Named third parties stand in for regulation the user cannot see; showing them at the moment of payment reduces perceived risk.	[VERIFY] Are Brink's / ICICI Lombard / Vistra shown on the payment screen or only on the website?
3	Vernacular, low-literacy-friendly interface.	<i>Onboarding friction / accessibility.</i> Native language and large numerals cut cognitive load for the tier-2/3 majority.	[VERIFY] Language picker at first launch? Is the whole flow translated, including KYC and errors?
4	Visible, reversible exit: sell anytime, or convert to coins or Nek jewellery.	<i>Loss aversion and reversibility.</i> Users commit more when undoing is easy and visible; delivery makes "digital" gold tangible.	[VERIFY] Taps to sell; time for money to reach the bank; where delivery options appear.
5	Round-ups and nudges that save spare change from everyday spends.	<i>Mental accounting / pain of paying.</i> Small amounts attached to spends that already happened feel costless; streaks and nudges reinforce the loop.	[VERIFY] Is round-up still offered? Is it SMS-based or native to Jar UPI (@jarunity)? Which streak or reward mechanics appear?

Screenshot slots 5.1-1 to 5.1-5: home screen with daily-saving card · payment screen with trust badges · language picker · sell/withdraw screen · round-up or streak screen

5.2 Five areas to improve

#	Improvement	Reasoning (UX principle)	What to check / propose
1	Show the all-in cost before purchase: 3% GST plus the buy-sell spread, as "if you sold right now you'd get ₹X."	<i>Information scent / honest pricing.</i> Third-party reviews report spreads of roughly 2.5–6% across digital gold apps. Discovering the cost on exit breaks trust. RBI and industry voices have called for standardised disclosure.	[VERIFY] Are buy price, sell price and GST shown together before payment?

#	Improvement	Reasoning (UX principle)	What to check / propose
2	A plain-language "what protects my gold" page.	<i>Trust repair after a public shock.</i> SEBI's Nov 2025 advisory said digital gold is outside its oversight. Users need a calm explainer: custody, audit certificate date, insurance, what happens if Jar fails.	[VERIFY] Is there an in-app explainer? Is it findable in two taps from the gold balance?
3	Withdrawal status and error recovery: a progress tracker with an ETA, and KYC prompted <i>before</i> the first withdrawal.	<i>Visibility of system status; help users recover from errors</i> (Nielsen). Public reviews cite slow withdrawals and surprise document requests at exit, which is the worst moment for friction.	[VERIFY] When is KYC asked? Is there a tracker and a support path inside the flow?
4	Pause, skip or lower the daily saving without cancelling the mandate.	<i>Habit loops need slack.</i> A user in a cash crunch who can only cancel churns for good; a seven-day pause preserves the habit and the mandate.	[VERIFY] Options shown when a user tries to stop AutoPay.
5	Goals with progress (Diwali, wedding, Akshaya Tritiya).	<i>Goal-gradient and endowed-progress effects.</i> Savings tied to a named goal and a progress bar keep going longer than an open-ended balance.	[VERIFY] Do goals exist? If yes, replace this with the weakest flow observed in the session.

Screenshot slots 5.2-1 to 5.2-5: pre-payment price breakdown · safety/FAQ page · withdrawal flow · AutoPay management · goals (or absence)

[VERIFY] Replace-if-wrong rule. If any "works well" item is broken in practice, move it to 5.2 with what you observed. Then promote the strongest item from your session notes (e.g., gamified rewards, gold-price alerts) into 5.1.

6. Q3 — Product exploration

6.1 Starting point: Jar's strengths and the category's open gap

Jar's moat is behavioural. It has (i) **automation**: UPI AutoPay mandates across more than 35 million registered users; (ii) **micro-ticket access** at ₹10; (iii) **UX simplicity**, in nine languages, for a base that is about 60% tier-2/3; and (iv) **established credibility with first-time savers**, backed by a named custodian (Brink's), insurer (ICICI Lombard) and trustee (Vistra).

The category has an open gap that every platform shares. SEBI's advisory of 8 November 2025 noted that digital gold is neither a security nor a regulated commodity derivative, so it sits outside SEBI's oversight. Industry and RBI voices have since called for standardised disclosure of storage, fees, redemption and insurance. Jar is best placed to lead here. It already publishes its custody, insurance and trustee partners, and it has the scale to set a disclosure standard others follow. Leading on transparency in digital gold, while extending its automation onto regulated rails, both deepens Jar's credibility and opens new revenue.

Start with regulated gold and goal-linked Nek savings; gate credit behind compliance

Analyst-judgement scores (1-5). Priority = impact ÷ effort



Figure 10. Effort vs impact for the five opportunities (analyst-judgement scores, 1–5).

Rank	Opportunity	Impact	Effort	Quadrant	Horizon
1	B. Goal-linked jewellery savings with Nek	3.5	1.5	Quick win / do now	Now (0-3 months)
2	A. Regulated gold SIP (ETF / FoF)	5.0	2.5	Quick win / do now	Now (0-6 months)
3	C. Insured micro-deposits via bank partners	4.0	3.5	Big bet / sequence	Next (6-12 months)
4	D. Sachet insurance on the AutoPay rail	3.5	3.5	Big bet / sequence	Next (9-15 months)
5	E. Gold-backed micro-credit (partner-led)	4.5	4.5	Big bet / sequence	Later (gated, 12+ months)

6.2 Opportunity cards

A. Regulated gold SIP (Gold ETF / fund-of-funds) — the strategic priority

- **User problem.** Savers who want gold *with* investor protection don't know how to buy an ETF.
- **Why Jar wins.** It already holds the AutoPay mandate and the gold-saving intent. A "Gold+ (SEBI-regulated)" option moves an existing habit rather than acquiring a new customer. Low-ticket mutual-fund SIPs (₹100–500 at many AMCs) fit Jar's micro-ticket DNA.
- **Integration path.** Become a mutual-fund distributor, or partner with a registered platform → add a second "jar" on the home card → route new savings there, with a cost comparison shown.
- **Monetisation.** Distribution trail on regular plans, or a platform fee on direct plans. Lower per-rupee take than the digital-gold spread, but recurring and defensible.
- **Sizing (back-of-envelope).** Say 5% of 35 million registered users (17.5 lakh) run a ₹500/month SIP. That is about ₹1,050 crore of regulated inflow in year one, At an indicative 0.3% annual trail, that book earns a year-end run-rate of about ₹3.15 crore a year. This is before market returns, churn and redemptions.
- **Key risk.** Cannibalising the digital-gold spread; distributor conduct rules (suitability, commission disclosure).
- **Success metric.** Share of active savers with a regulated gold SIP within 6 months; regulated share of gold AUM.

B. Goal-linked jewellery savings with Nek — the quick win

- **User problem.** Households save for jewellery around weddings, Akshaya Tritiya and Dhanteras, often through jeweller 11-month schemes that lock money to one store.
- **Why Jar wins.** Nek is Jar's own jewellery brand. Automation plus a named goal ("Wedding necklace, ₹45,000 by Feb") turns open-ended saving into a purchase the user can see coming.
- **Integration path.** Create a goal on the home card → progress bar with a projected date → one-tap redemption at Nek, with a making-charge discount for completed goals.
- **Monetisation.** Nek margin and making charges. CAC is lower because the buyer is pre-funded and pre-committed.
- **Sizing (back-of-envelope).** Say 2% of registered users (7 lakh) set a jewellery goal averaging ₹10,000, and 30% complete and redeem at Nek. That gives about 2.1 lakh orders and ₹210 crore of GMV, roughly 2.1× Nek's reported annual revenue of over ₹100 crore.
- **Key risk.** Must be framed as accumulation of the user's own gold, not a deposit scheme. Fulfilment quality at Nek.
- **Success metric.** Goal completion rate; share of completed goals redeemed at Nek; 12-month retention of goal vs non-goal savers.

C. Insured micro-deposits via bank partners

- **User problem.** First-time savers want a safe, fixed-return home for emergency money, and bank FDs feel out of reach.
- **Why Jar wins.** The same mandate rail can sweep daily ₹10s into an RD or FD at partner small finance banks. Jar already works with Unity Small Finance Bank for UPI. DICGC insurance up to ₹5 lakh per depositor per bank is a regulated trust anchor Jar can explain simply.
- **Integration path.** Bank-partner FD/RD marketplace → a "safe jar" next to the gold jar → auto-split rules (e.g., 70% gold, 30% deposit).
- **Monetisation.** Distribution fee from partner banks; cross-sell lift.

- **Key risk.** Partner concentration; outsourcing and KYC obligations; keeping deposit messaging separate from gold.
- **Success metric.** Deposit balance per active saver; share of savers holding both gold and a deposit.

D. Sachet insurance on the AutoPay rail

- **User problem.** Tier-2/3 and gig workers are under-insured. Annual premiums feel large; ₹2–5 a day does not.
- **Why Jar wins.** Daily auto-debit is exactly how sachet premiums need to be collected, and Jar's plain-language style suits explaining cover.
- **Integration path.** IRDAI corporate-agent licence or broker partner → add-on toggles (personal accident, hospital cash) when a saving streak is set up.
- **Monetisation.** Commission within IRDAI expense limits.
- **Key risk.** Mis-selling or claim rejections would erode the core trust asset.
- **Success metric.** Attach rate; claim turnaround; 13-month persistency.

E. Gold-backed micro-credit (partner-led) — gated

- **User problem.** Small emergencies are funded by informal lenders, or by selling gold at a bad moment.
- **Why Jar wins.** The collateral is already vaulted and valued daily, so a small loan needs no branch visit or assayer.
- **Integration path.** Jar as lending service provider to an RBI-regulated bank or NBFC → "borrow instead of sell" offered on the sell screen.
- **Monetisation.** Sourcing fee or revenue share on disbursements.
- **Key risk.** The highest regulatory load of the five: digital-lending and gold-loan rules, plus the open question of pledging *digital* gold as collateral. Start only after the regulated rails (A, C) are live.
- **Success metric.** Loans per 1,000 active savers; 90+ day delinquency; share of borrowers who keep saving.

Considered and deprioritised. Digital silver adds the same regulatory exposure as gold for little new value. B2B payroll-linked saving has long enterprise sales cycles; revisit after A–C.

6.3 Sequenced roadmap

Horizon	Moves	Milestone
0–6 months	B (Nek goals) and A (regulated gold SIP); publish a digital-gold disclosure standard	Regulated product live; first goal cohorts complete
6–15 months	C (insured deposits), then D (sachet insurance)	Majority of new savings on regulated or insured rails
12+ months, gated	E (gold-backed credit) via a regulated partner	Clean compliance; delinquency within plan

6.4 Sources (Sections 5–6)

TechCrunch, "Indian fintech Jar turns profitable by helping millions save in gold", 18 Sep 2025: users, tier-2/3 share, languages, Nek revenue. · Jar website (myjar.app): saving modes and minimums; custody, insurance and trustee partners. Accessed Oct 2026. · Inc42, "India's Digital Gold Rush Gets Regulatory Reality Check", 12 Nov 2025; Groww explainer: SEBI advisory of 8 Nov 2025 and calls for standardised disclosure. · Entrackr: Jar enters UPI payments through BharatPe and Unity Small Finance Bank, 2025. · Third-party reviews and spread comparisons (IPOMarket, Jupiter, Analytics Insight): directional only.

Appendix A — Assumptions log

#	Assumption	Why	Effect if wrong
A1	Order Date datetime cells have day and month swapped; text cells are dd-mm-yyyy	Only reading that makes dates monotonic in sequential Order ID	Part 2 monthly actuals would shift months
A2	Sales Target "day" component encodes the year (18 → 2018)	Months then run Apr-2018 to Mar-2019, exactly the order period	No overlap for target-vs-actual
A3	Target values are monthly <i>sales</i> (Amount) targets in ₹	Same order of magnitude as monthly category sales	Attainment metrics would not apply
A4	"Average profit per order" = profit ÷ distinct orders containing the group	Matches the brief's wording; per-line reported alongside	Magnitudes change; rankings do not
A5	Lower realised unit price on loss lines may reflect discounting (hypothesis)	No discount column exists	Product mix explains the same pattern; loss concentration still holds
A6	3 "Delhi, Madhya Pradesh" orders kept as recorded	Brief analyses the State field	Madhya Pradesh loses 3 orders; still #1
A7	Re-phased target uses same-year, company-wide seasonality, which includes Furniture	Only one year of data	MAPE gain is an upper bound; ex-Furniture sensitivity: 27.3%
A8	Q3 impact/effort scores and sizing inputs are analyst judgement	No internal Jar data	Ranking and sizes are a starting point for discussion

Appendix B — Detail tables

B1. Sub-category performance

Category	Sub-category	Sales	Profit	Margin	Lines	Orders	Loss lines	Profit / line
Clothing	Trousers	₹30,039	₹2,847	9.5%	39	37	38.5%	₹73.00
Clothing	Stole	₹18,546	₹2,559	13.8%	192	157	27.1%	₹13.33
Clothing	Hankerchief	₹14,608	₹2,098	14.4%	198	138	21.2%	₹10.60
Clothing	T-shirt	₹7,382	₹1,500	20.3%	77	70	15.6%	₹19.48
Clothing	Shirt	₹7,555	₹1,131	15.0%	69	66	17.4%	₹16.39
Clothing	Saree	₹53,511	₹352	0.7%	210	156	46.7%	₹1.68
Clothing	Leggings	₹2,106	₹260	12.3%	53	49	30.2%	₹4.91
Clothing	Skirt	₹1,946	₹235	12.1%	64	56	28.1%	₹3.67
Clothing	Kurti	₹3,361	₹181	5.4%	47	41	29.8%	₹3.85
Electronics	Printers	₹58,252	₹5,964	10.2%	74	67	33.8%	₹80.59
Electronics	Accessories	₹21,728	₹3,559	16.4%	72	65	22.2%	₹49.43
Electronics	Phones	₹46,119	₹2,207	4.8%	83	71	49.4%	₹26.59
Electronics	Electronic Games	₹39,168	-₹1,236	-3.2%	79	73	50.6%	-₹15.65
Furniture	Bookcases	₹56,861	₹4,888	8.6%	79	72	27.8%	₹61.87
Furniture	Furnishings	₹13,484	₹844	6.3%	73	66	41.1%	₹11.56
Furniture	Chairs	₹34,222	₹577	1.7%	74	64	52.7%	₹7.80
Furniture	Tables	₹22,614	-₹4,011	-17.7%	17	16	64.7%	-₹235.94

B2. Furniture: target vs actual by month

Month	Target	Actual	Gap	Attainment	Furniture profit	Re-phased target
Apr 2018	₹10,400	₹8,121	-₹2,279	78.1%	-₹3,425	₹10,079
May 2018	₹10,500	₹6,220	-₹4,280	59.2%	-₹794	₹8,792
Jun 2018	₹10,600	₹5,532	-₹5,068	52.2%	-₹856	₹7,287
Jul 2018	₹10,800	₹3,483	-₹7,317	32.2%	-₹457	₹3,993
Aug 2018	₹10,900	₹9,538	-₹1,362	87.5%	₹443	₹9,517
Sep 2018	₹11,000	₹8,704	-₹2,296	79.1%	-₹2,468	₹8,201
Oct 2018	₹11,100	₹6,766	-₹4,334	61.0%	-₹1,316	₹9,737
Nov 2018	₹11,300	₹15,165	₹3,865	134.2%	₹3,945	₹14,810
Dec 2018	₹11,400	₹9,474	-₹1,926	83.1%	₹187	₹11,574
Jan 2019	₹11,500	₹21,257	₹9,757	184.8%	₹3,284	₹18,923
Feb 2019	₹11,600	₹16,262	₹4,662	140.2%	₹2,168	₹11,834
Mar 2019	₹11,800	₹16,659	₹4,859	141.2%	₹1,587	₹18,152

B3. All states

Rank	State	Orders	Sales	Profit	Margin	Profit / order	Sales share	Profit share
1	Madhya Pradesh	101	₹1,05,140	₹5,551	5.3%	₹54.96	24.4%	23.2%
2	Maharashtra	90	₹95,348	₹6,176	6.5%	₹68.62	22.1%	25.8%
3	Rajasthan	32	₹21,149	₹1,257	5.9%	₹39.28	4.9%	5.2%
4	Gujarat	27	₹21,058	₹465	2.2%	₹17.22	4.9%	1.9%
5	Punjab	25	₹16,786	-₹609	-3.6%	-₹24.36	3.9%	-2.5%
6	Delhi	22	₹22,531	₹2,987	13.3%	₹135.77	5.2%	12.5%
7	Uttar Pradesh	22	₹22,359	₹3,237	14.5%	₹147.14	5.2%	13.5%
8	West Bengal	22	₹14,086	₹2,500	17.7%	₹113.64	3.3%	10.4%
9	Karnataka	21	₹15,058	₹645	4.3%	₹30.71	3.5%	2.7%
10	Kerala	16	₹13,459	₹1,871	13.9%	₹116.94	3.1%	7.8%
11	Bihar	16	₹12,943	-₹321	-2.5%	-₹20.06	3.0%	-1.3%
12	Andhra Pradesh	15	₹13,256	-₹496	-3.7%	-₹33.07	3.1%	-2.1%
13	Nagaland	15	₹11,903	₹148	1.2%	₹9.87	2.8%	0.6%
14	Jammu and Kashmir	14	₹10,829	₹8	0.1%	₹0.57	2.5%	0.0%
15	Haryana	14	₹8,863	₹1,325	14.9%	₹94.64	2.1%	5.5%
16	Himachal Pradesh	14	₹8,666	₹656	7.6%	₹46.86	2.0%	2.7%
17	Goa	14	₹6,705	₹370	5.5%	₹26.43	1.6%	1.5%
18	Sikkim	12	₹5,276	₹401	7.6%	₹33.42	1.2%	1.7%
19	Tamil Nadu	8	₹6,087	-₹2,216	-36.4%	-₹277.00	1.4%	-9.3%

B4. Cities in the top-5 and loss-making states

State	City	Orders	Sales	Profit	Margin	Share of state sales	Loss lines
Andhra Pradesh	Hyderabad	15	₹13,256	-₹496	-3.7%	100.0%	26.2%
Bihar	Patna	16	₹12,943	-₹321	-2.5%	100.0%	38.7%
Gujarat	Ahmedabad	17	₹14,230	-₹880	-6.2%	67.6%	40.3%
Gujarat	Surat	10	₹6,828	₹1,345	19.7%	32.4%	8.0%
Madhya Pradesh	Indore	76	₹79,069	₹4,159	5.3%	75.2%	36.0%
Madhya Pradesh	Bhopal	22	₹23,583	₹871	3.7%	22.4%	39.4%
Madhya Pradesh	Delhi	3	₹2,488	₹521	20.9%	2.4%	0.0%
Maharashtra	Mumbai	68	₹61,867	₹1,637	2.6%	64.9%	41.1%
Maharashtra	Pune	22	₹33,481	₹4,539	13.6%	35.1%	28.9%
Punjab	Chandigarh	16	₹12,279	-₹1,153	-9.4%	73.2%	42.2%
Punjab	Amritsar	9	₹4,507	₹544	12.1%	26.8%	13.3%
Rajasthan	Udaipur	13	₹11,073	₹2,010	18.2%	52.4%	10.0%

State	City	Orders	Sales	Profit	Margin	Share of state sales	Loss lines
Rajasthan	Jaipur	19	₹10,076	-₹753	-7.5%	47.6%	27.3%
Tamil Nadu	Chennai	8	₹6,087	-₹2,216	-36.4%	100.0%	40.0%

Appendix C — Reproducibility

- **Run:** `python main.py`. This regenerates every table in `outputs/tables/`, every figure in `outputs/figures/`, `outputs/report_numbers.json`, the report Markdown and HTML in `report/`, and this PDF. Every number in this report is injected from `report_numbers.json`, and the build fails on any unresolved placeholder or broken claim. The Q1 analysis is also walked through step by step in `Q1_Sales_Analysis.ipynb`.
- **Environment:** Python 3.12.1; pandas 3.0.6; matplotlib 3.11.2; PDF rendered by headless Chromium.
- **Data:** raw Excel files in `data/` are read-only; all repairs happen in memory (Section 1.2).

Appendix D — Q2 hands-on checklist (≈ 20 minutes)

1. **Install → first saving (time it).** Count taps and screens to the first AutoPay mandate. Note the default amount and any language prompt. (5.1 #1, #3)
2. **Payment screen.** Screenshot the trust badges shown, if any. Is the buy price split into gold value, GST and spread? (5.1 #2, 5.2 #1)
3. **Sell ₹10 worth.** Note the sell price vs buy price, taps, KYC prompts, time for money to reach the bank, and status messages. (5.1 #4, 5.2 #3)
4. **Try to stop daily saving.** Are pause, skip or lower options offered before cancel? (5.2 #4)
5. **Search for safety information.** From the gold balance, how many taps to custody, audit and insurance details? Any mention of the SEBI advisory? (5.2 #2)
6. **Explore engagement.** Round-ups, streaks, spins and rewards, goals, notifications on day 1. (5.1 #5, 5.2 #5)
7. **Cross-sell surfaces.** Where do Nek, UPI, loans or insurance appear? Do they crowd out the saving card?
8. **Replace or confirm.** For each [VERIFY] row, mark ✓ confirmed / ✎ changed / X replaced, and drop the screenshot into its slot.